

Review Article

Impact of climate change on global economy: A comprehensive review

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ABSTRACT

Climate change has emerged as an important global issue, primarily driven by human activities, affecting ecosystems, societies, and economies worldwide. This review employed a comprehensive narrative analysis that incorporates expertise from policy studies, environmental science, and economics. Targeted keywords are used to systematically search major academic databases. This article explores various aspects of climate change, economic consequences, and the evolving discourse on climate risk. It addresses the association between climate change and several economic variables, such as gross domestic product (GDP), foreign direct investment (FDI), and financial markets. The review highlights the complex relationship among physical risks, transition risks, and approaches to cope with climate change. It examines global climate policies such as the Kyoto Protocol and Paris Agreement. This study also investigates climate risk assessment models like Integrated Assessment Models (IAMs) and how they help understand the economic effect of climate change. Moreover, it explores ways to adapt to and reduce climate change, such as carbon taxes and subsidies. This review focuses on the critical importance of instruments like the Climate Action Tracker in evaluating national and global climate change responses, which serve as a framework for assessing policy effectiveness and progress toward emission targets. By integrating different viewpoints from economics, environmental science, and governance, this review underscores the value of coordinated global action to address the challenges posed by climate change.

1. Introduction

Climate change, primarily propelled by human activities, has arisen as a paramount challenge of the contemporary era, significantly impacting ecosystems, societies, and economies at a global scale (Bandh et al.,

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2021). Climate change is a complex global challenge that affects many aspects of the environment, society, and economy requiring cooperation among governments worldwide (Adger et al., 2005; Leal Filho et al., 2021; Feliciano et al., 2022). While climate change encompasses a broader range of phenomena beyond temperature increases, denoting both a collection of physical occurrences and a focal point in public policy discussions (Weber and Stern, 2011). Several factors contribute to this phenomenon, such as changes in the Earth's energy balance, fluctuations in snow and ice coverage, and variations in oceanic conditions (Benestad, 2006). These changes observed are a result of human activities, specifically the emission of greenhouse gases (GHGs) such as carbon dioxide (CO₂) and methane (CH₄) (Eggleton, 2012). The effects of climate change are substantial, with potential future outcomes, such as severe weather events, increased sea levels, and disturbances to ecosystems (Ummenhofer and Meehl, 2017). Tackling climate change demands a comprehensive strategy that encompasses both individual and collective efforts, along with the implementation of impactful policy measures (Cheshmehzangi and Dawodu, 2018). Globally, climate change is the phrase employed to describe persistent shifts in precipitation, temperatures, and weather patterns driving by human-caused GHG emissions (Kumar et al., 2021). Rising sea levels in the Pacific Region and more intense heatwaves in South Asia are only two examples of how climate change manifests regionally. Locally, climate change manifests itself as specific phenomena that will have a direct impact on livelihoods, which include unseasonable rains, crop failures, or persistent drought.

Climate extremes should be distinguished from climate change. Climate extremes are short-term, high intensity phenomena, such as floods, droughts, storms, and heatwaves, whereas climate change is a long-term shift in the average climate (Valjarević et al., 2025). These extremes have a significant impact on economies, particularly in developing countries where institutional infrastructure and resources may be lacking and are frequently accelerated by underlying climate change (Clarke et al., 2022).

Since 1850, the Earth's temperature has increased at an average rate of 0.06°C per decade. However, since 1982, the rate of warming has accelerated significantly, reaching 0.20°C per decade, which is more than 3 times faster than the long-term average (Intergovernmental Panel on Climate Change (IPCC), 2014a; Lindsey and Dahlman, 2025). In 2023, the Earth experienced its highest temperature on record since reliable global monitoring began in 1850, surpassing previous records by a significant margin (Dunn et al., 2023; Zheng et al., 2023; Li et al., 2024a). The temperature was 1.18°C higher than the average temperature of 13.00°C in the 20th century (NCEI, 2018). During 2014–2023, the historical record indicated that the Earth has experienced its highest temperature ever recorded, marking a notable trend toward increasing warmth (Li et al., 2024b).

Recent changes in annual surface temperatures from 1994 to 2023 show a sharper increase compared to the trend observed since the early 1900s (Wu et al., 2023). The current warming trend is notably faster than the average over the longer period. In certain areas, temperatures have risen by 1.00°C or more per decade (Song et al., 2022). The most significant variations are observed in the Arctic, where the melting of ice and snow, which usually reflect sunlight, accelerates the warming process (You et al., 2021).

2. Methodology

This article synthesizes interdisciplinary research on the financial impact of climate change using comprehensive narrative review methodology. The review incorporates findings from risk analysis, policy studies, economics, and environmental science, with an in-depth comprehension of how climate change disrupts financial and macroeconomic systems and how global governance mechanisms resolve these issues.

Peer-reviewed scholarly publications, policy reports, and international databases were thoroughly reviewed. Major scholarly databases such as Google Scholar, Web of Science, ScienceDirect, and Scopus were used to conduct the search. Climate change, economic impact, gross domestic product (GDP) and climate, foreign direct investment (FDI) and climate, climate risk, financial markets and climate, climate policy, Integrated Assessment Models (IAMs), and Climate Action Tracker were the keywords utilized. While important earlier studies (e.g., Nordhaus, 1991) were also included, articles published between 2000 and 2024 were given priority to maintain current relevance. Global temperature anomalies in this review were computed by subtracting the long-term average temperature (usually the baseline period of 1951–1980) from the annual global temperature.

Economists have long been discussing how climate change affects the overall economy. William Nordhaus, a pioneer in analyzing the economic implications of cutting carbon emissions, recently cautioned that our current efforts to address global warming may not be sufficient (Nordhaus, 2018). A broader group of economic experts comprehends the economic impacts of climate change, containing those involved in macroeconomic modelling and prediction across national governments, central banks, international organizations, and private institutions (Batten,

2018). The impacts of climate change are becoming more evident. Recent research suggests that advanced financial markets, including debt, stocks, and real estate, have considered the risks posed by climate change. For example, analysis across various regions indicates that indicators of climate change are significantly linked to changes in sovereign debt yields (Volz et al., 2020; Beirne et al., 2021; Boitan and Marchewka-Bartkowiak, 2022; Cevik and Jalles, 2022; Kling et al., 2025).

The conceptual framework (Fig. 1) presented in this review utilizes a variety of perspectives from different fields, such as economics, environmental science, sociology, and others. It investigates how climate change affects economic systems, including both direct impacts on physical assets and natural resources, as well as broader economic indicators like GDP, FDI, and financial markets. Moreover, the article examines how different economic sectors are affected by climate change, focusing on their varying levels of vulnerability. The Climate Action Tracker, a global benchmarking tool that provides standard metrics for assessing national climate goals and actions, is an essential aspect of this study, which uses economic assessment models for evaluating climate adaptation and mitigation strategies.

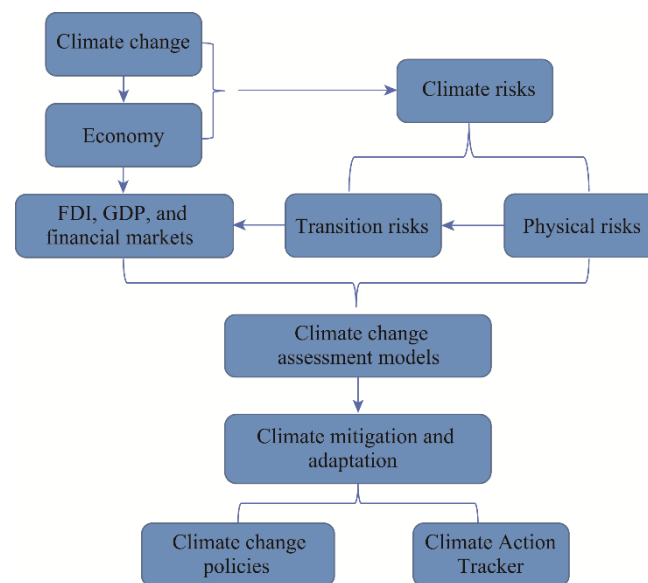


Fig. 1. Conceptual framework of this study. FDI, foreign direct investment; GDP, gross domestic product.

3. Results and discussion

3.1. Economic impact of climate change

Climate change has significant implications for the global economy, potentially impacting agriculture, economic growth, and development (FitzRoy, 2016; Popkova, 2022). These effects are particularly pronounced in developing countries, where a win-win approach to climate policy that prioritizes sustainable development is crucial (Reddy, 2009). International climate change law plays a key role in addressing these challenges, focusing on environmental, economic, and ethical perspectives (Bodansky, 2017). However, the specific economic consequences of climate change and the potential benefits of mitigation are complex and multifaceted (Olaru, 2018).

Climate change initially improves economic welfare but later in the century, with predominantly negative impacts, affecting income and biodiversity (Tol, 2009). Climate change has a direct impact on economic aspects like agricultural production and essential resources such as water and public health. Additionally, climate change can also indirectly influence various economic sectors, such as manufacture, energy generation, transportation, and other services (Arent et al., 2015). In many studies examining the economic consequences of global warming, researchers evaluated and quantified the impacts of climate change individually for different sectors. These assessments are then combined to estimate the overall change in social welfare (Nordhaus, 1991; Cline, 1992; Tol, 1995; Neumann and Mendelsohn, 1999). This is called the enumerative approach.

The need for coordinated action in response to climate change is underscored by the United Nations' Sustainable Development Goals (SDGs), particularly SDG 13 (climate action) (Popkova, 2022). The economic costs of climate

change and the benefits of mitigation efforts are key considerations in this context (Popkova, 2022). The global response to climate change, as explored in a range of countries and by the International Energy Agency, is a complex and multifaceted issue. The specific impacts of climate change on developing countries, particularly in sub-Saharan Africa and South-East Asia, are also a focus of research (Akpalu, 2015).

If the global temperature rises by 1.92°C by 2050, the world's GDP could decrease by about 0.50% compared to a hypothetical scenario with no climate change (Bosello et al., 2012). Additional investigations reveal that the influence of climate change on the global annual GDP is expected to amplify progressively, resulting in a reduction in global GDP ranging from 0.70% to 2.50% by 2060 based on the most probable climate sensitivity range (Dellink et al., 2014). These global annual GDP reductions significantly conceal larger discrepancies across various sectors and regions. While Northern Europe may experience a slight positive impact (0.18%), Southern and Eastern Europe are projected to face adverse effects (−0.15% and −0.21%, respectively) due to climate change. The most at risk are less developed regions, such as South Asia, South-East Asia, North Africa, and Sub-Saharan Africa (Bosello et al., 2012).

Acharya et al. (2022) found that increased local exposure of GDP to heat stress correlates with wider spreads on municipal bonds. This is primarily due to anticipated rises in energy costs and declines in labor productivity. Regarding Standard & Poor's 500 companies, the researchers observed that greater exposure to heat stress leads to higher expected returns across all stocks (Ardia et al., 2023). However, yields only rise for sub-investment grade corporate bonds, not for investment grade bonds. These effects became significant from 2013 to 2015, specifically concerning heat stress exposure rather than other physical risks (Acharya et al., 2022).

Paris Agreement has the potential to lead to immense global economic benefits, estimated at around 174.89×10^{12} USD/a in the foreseeable future (Kompas et al., 2018). Figure 2 illustrates the interrelationship among global annual GDP, CO₂ emissions, and population changes.

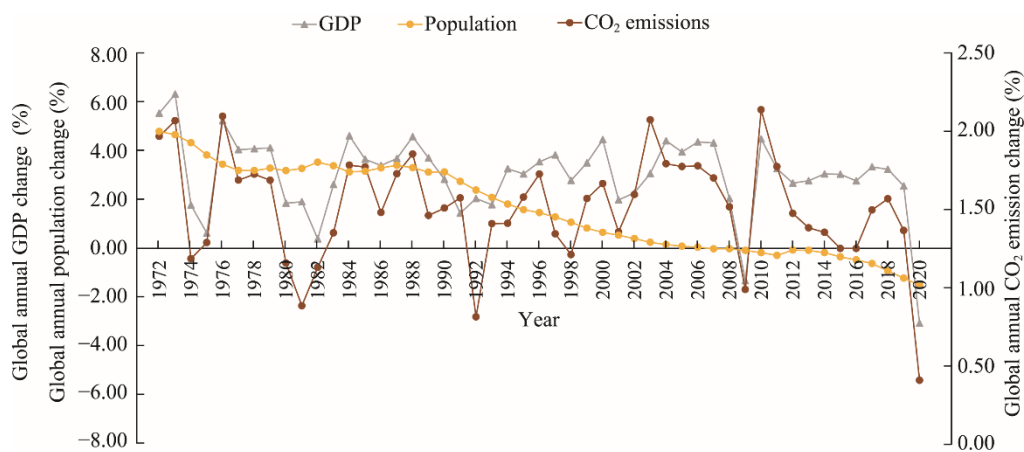


Fig. 2. Global annual GDP, population, and carbon dioxide (CO₂) emission changes during 1972–2020. The data were from the United Nations, Department of Economic and Social Affairs, Population Division (2022) and Friedlingstein et al. (2023).

Climate change presents substantial risks to the financial sector, potentially affecting asset values and influencing investment strategies. The economy is influenced by climate change, and countries with high emissions and significant annual climate variations face increased financial risks in markets (Monasterolo, 2020). Unlike advanced economies, they are penalized with a risk premium. Emerging market economies need to spend more on public initiatives to adapt to climate change and build resilience, despite already dealing with high levels of debt (Hallegatte et al., 2020). Emerging evidence suggests that financial markets are considering risks associated with climate change. However, various institutional barriers hinder the complete pricing of these risks.

In a potential worst-case scenario for climate impact, it is projected that by 2030, there could be a significant reduction in portfolio value, ranging from 5.00% to 20.00%, compared to its value without the effects of global warming. However, this risk can be significantly mitigated through swift action towards transitioning to cleaner energy sources, thereby reducing GHG emissions (Covington and Thamotheram, 2015). Further, long-term investors should actively promote a swift transition to sustainable energy practices, and investment managers may be obliged to support this transition to align with their fiduciary duties (Covington and Thamotheram, 2015).

The impact of temperature extremes, specifically those occurring at low frequencies, has a notable adverse effect

on equity valuations. From an economic perspective, an increase of 1.00°C in temperature correlates with approximately a 5.00% reduction in equity valuation (Bansal et al., 2015). The transition to a low-carbon economy is expected to have a significant impact on market value. Historically, long-term carbon goals were not factored into market assessments, leading to excessive investments in fossil fuels. However, these investments may turn into stranded assets due to evolving policy, market dynamics, technological advancements, and social shifts (Robins, 2014). Now, people rely heavily on fossil fuel energy. Introducing a carbon price could lead to various indirect consequences across all sectors, ultimately impacting the entire economy. When the carbon tax reaches significant levels, the entire economy will probably feel its effect. Consequently, carbon regulations pose a significant risk in financial markets (Caldecott and McDaniels, 2014).

The exploration of managing uncertainty in climate through financial decisions has not received widespread attention, though there are a few notable studies such as those by Andersson et al. (2016) and Engle et al. (2020). These studies suggest methods for creating investment portfolios with reduced carbon exposure using readily available stocks. The field of financial markets is where these discussions primarily occur. In climate uncertainty, green bonds could provide sturdy and secure returns. They provide an alternate perspective on how these assets could potentially be employed not only as socially responsible investments but also for overseeing climate-related risks in investment portfolios through financial markets (Cepni et al., 2020).

3.2. Climate change and global economic flows

In the long term, temperature fluctuations affect FDI flows differently, depending on a country's level of economic development. Developing countries tend to experience a decline in FDI flows due to rising temperatures, while developed countries may see an increase in FDI flows. This observation presents a new perspective for understanding how to finance climate change mitigation and adaptation efforts. Additionally, it provides valuable insights for reconsidering global FDI policies in the context of climate change (Barua et al., 2020). Sarkodie (2020) emphasized the role of renewable energy in combating climate change, noting that while factors such as FDI, economic development, and governance can exacerbate climate-related challenges, the growing adoption of renewable energy can help mitigate these effects. These findings collectively highlight the complex relationship between climate change and FDI, showing its dual potential to contribute to augmenting and alleviating climate-related issues.

Multinational corporations (MNCs) develop global investment strategies with heightened caution due to the growing uncertainties posed by climate change. Typically, MNCs prefer regions with lower risks and higher returns (Shear et al., 2023). According to the Ownership, Location, Internalization (OLI) framework, FDI tends to decrease in regions where perceived risks are high (Dunning, 1977). Recent studies confirm that institutional or policy risks (such as political instability, corruption, or regulatory unpredictability) at the national level are linked to reduced FDI inflows (Cheng and Kwan, 2000; Leahy and Montagna, 2001; Bjorvatn and Eckel, 2006; Alfaro et al., 2008; Julio and Yook, 2016; Okafor et al., 2022). In today's global economy, country-specific risks have become increasingly significant, with trade tensions, armed conflicts, terrorism, weak governance, and corruption all exerting a growing influence on international business operations (Cavusgil et al., 2020).

China is increasingly playing a leading role in global foreign investment (Yan et al., 2023). Li and Gallagher (2022) indicated that Chinese abroad investments face greater exposure to physical climate risks including water stress, flooding, hurricanes, and typhoons compared to those from other countries. However, the climate risk levels for Chinese facilities are generally like other countries. Over time, the increasing presence of FDI in underdeveloped countries has positive effects on the scope, intensity, and environmental efficiency of CO₂ emissions (Mejia, 2023). Emerging markets and developing economies (EMDEs) with higher exposure to climate-related risks tend to receive lower levels of FDI. However, EMDEs with more developed financial systems are often better equipped to counteract the negative effects of climate risk on investment inflows (Gopalan et al., 2023).

3.3. Managing economic risks and transitioning to sustainability

Climate risk constitutes a major threat to human well-being and economic stability, attracting global attention due to its far-reaching effects on health, livelihoods, and financial systems (Xing and Wang, 2023). Huang et al. (2018) investigated how climate-related risks influence the financing decisions of publicly listed firms across the globe, employing the Global Climate Risk Index developed by Germanwatch (Kreft and Eckstein, 2013) as a central

tool for analysis. Increasingly, research is focusing on how environmental conditions shape economic outcomes, particularly in relation to production, investment, and productivity (Batten, 2018).

To comprehensively understand the notion of “risk”, it is essential to break it down into its core components: hazard, probability, and vulnerability (or exposure). Climate vulnerability refers to the economic consequences of climate hazards, measuring the extent to which economies are financially exposed to these threats. Jones and Boer (2005) focused on specific climate hazards that imperil economic systems, conceptualizing vulnerability through their economic ramifications.

Groundbreaking research by the Bank of England (2015) and Carney (2015) introduced a framework to better understand climate-related financial risks. This framework classifies these risks into two main categories (as shown in Figure 3).

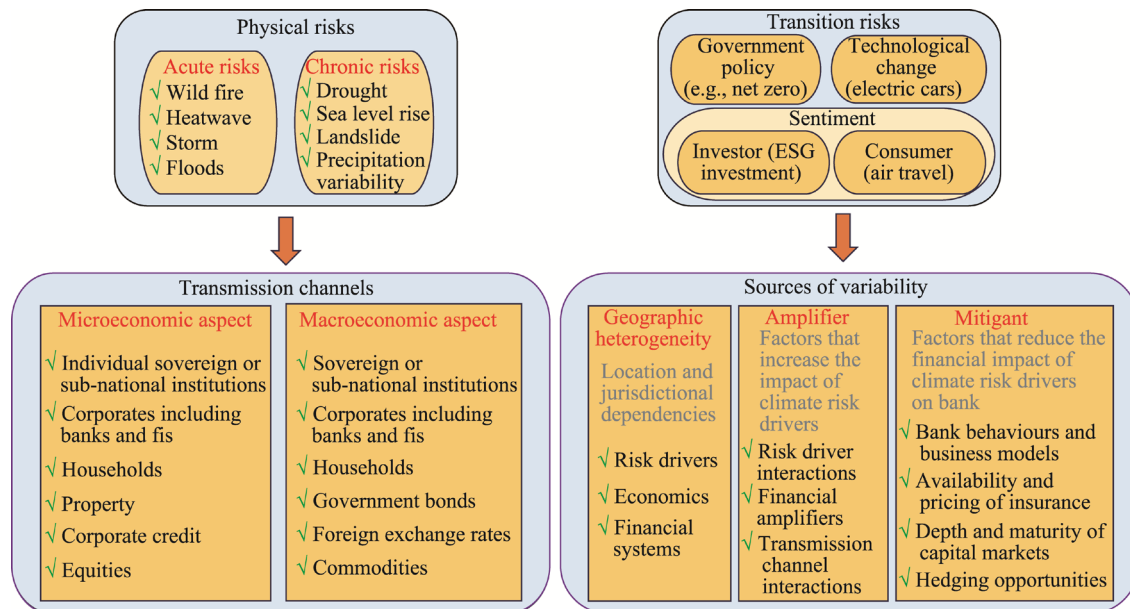


Fig. 3. Main types of climate risks and their transmission channels. ESG, Environmental, Social, and Governance.

Identifying and understanding the types of risks arising from climate change is essential for economic resilience and policy planning. These risks are generally categorized into three major types: (1) physical risks, which involve direct impacts from increasingly frequent and severe weather events (e.g., floods, hurricanes, and droughts), as well as long-term environmental changes such as rising sea levels and desertification; (2) transition risks, which stem from the policy, regulatory, market, and technological adjustments required to shift toward a more sustainable, low-carbon economy; and (3) liability risks, which pertain to potential future legal actions seeking compensation for damages caused by climate change, particularly targeting entities perceived to be responsible for emissions or inaction (Martin, 2020).

Zhang (2022) observed that economic systems in underdeveloped countries exhibit a weaker response to climate-related risks compared to those in developed economies. This disparity is largely attributed to the lack of public awareness, insufficient institutional capacity, and the absence of targeted climate policies. Further emphasizing the importance of institutional quality, Abdelzaher et al. (2020) found that countries characterized by higher levels of innovation, greater international economic openness, and robust regulatory environments tend to be more resilient to the adverse effects of climate change. Moreover, countries that demonstrate excellence in Environmental, Social, and Governance (ESG) standards often benefit from reduced sovereign default risks and narrower bond spreads, signaling lower perceived risk in international financial markets (Crifo et al., 2017; Capelle-Blancard et al., 2019).

3.3.1. Evolution of climate risks

The global average surface temperature has increased by approximately 0.87°C since 1900, marking a significant shift in the Earth’s climate system. This warming trend is projected to continue, bringing about a series of far-reaching environmental, economic, and social consequences that will affect all countries (Basel Committee on Banking Supervision, 2021).

Scientific evidence consistently attributes this temperature rise primarily to the accumulation of GHGs in the atmosphere because of human activities. The IPCC asserts with high confidence that elevated GHG concentrations are the most probable cause of global warming (IPCC, 2014b).

Moreover, the thermal effects of anthropogenic emissions are expected to persist for centuries or even millennia, especially if mitigation efforts remain insufficient. Without significant interventions, the warming already set in motion will continue to unfold over time, affecting generations to come (IPCC, 2018). Therefore, scientists and policy experts emphasize the necessity of transitioning toward a low-carbon economy, which is characterized by reducing reliance on fossil fuels and focusing on sustainable energy alternatives (IPCC, 2014a). However, this transition is not without risk. The transition to a low-carbon economy may carry potential threats of structural disruption, particularly if this transformation is driven by abrupt or uncoordinated policy changes, technological upheaval, or volatile shifts in market and consumer behavior. Sectors unprepared for this shift may face economic shocks, underscoring the critical need for strategic foresight and institutional resilience (Basel Committee on Banking Supervision, 2021).

3.3.2. Physical risks

Physical risks refer to the direct threats posed by climate change to human and natural systems, particularly through increased frequency and severity of extreme weather events and long-term environmental shifts (Martin, 2020). These risks are commonly classified into two categories: acute risks and chronic risks. Acute risks arise from extreme, sudden-onset events such as heatwaves, floods, wildfires, hurricanes, and cyclones. Chronic risks develop gradually over time, including sea level rise, desertification, ocean acidification, and sustained temperature increases (NGFS, 2019; McKinsey Global Institute, 2020).

While vulnerability to these risks is influenced by human activities such as land use changes, infrastructure planning, and socioeconomic conditions, their exact timing, frequency, and magnitude remain difficult to predict, complicating risk management efforts (Basel Committee on Banking Supervision, 2021). From 1980 to 2018, natural disasters caused an estimated economic loss of 5.20×10^{12} USD (Munich Re, 2020). Scholarly models and scenarios provide robust projections of the economic implications of these risks (Burke et al., 2015; Dietz et al., 2016; NGFS, 2020).

Acute risks include life-threatening natural disasters, such as intense heatwaves, flash floods and coastal inundation, severe tropical storms (hurricanes, cyclones, and typhoons), and wildfires driven by dry conditions and elevated temperatures (Clarke et al., 2022). Studies confirm that rising global temperatures are associated with increased wildfire frequency and intensity, which damages ecosystems, threatens biodiversity, and disrupts local economies (Abatzoglou and Williams, 2016; Abatzoglou et al., 2019; Jones et al., 2020).

Chronic risks evolve over longer timescales and include rising sea levels from glacial melt and thermal expansion as well as ocean acidification, affecting marine biodiversity. Persistent warming contributes to desertification and disrupts agricultural productivity (Greenhill et al., 2020; Han et al., 2021; Gampe et al., 2021).

These long-term shifts can lead to forced migration, resource scarcity, and humanitarian crises. Thalheimer and Webersik (2020) linked such risks to rising geopolitical instability. The World Bank (2018) estimated that by 2050, around 1.40×10^8 people in Latin America, South Asia, and Sub-Saharan Africa may be internally displaced due to water shortages and decreased agricultural yields. Furthermore, the IPCC (2019) warned that accelerated ice sheet melting and glacier retreat would likely elevate sea levels, heightening the risk of coastal flooding and displacement for millions of people in vulnerable regions.

3.3.3. Transition risks

Transition risks arise from the economic and financial uncertainties associated with shifting towards a low-carbon, environmentally sustainable economy. These risks are driven by changes in public policy, technological innovation, and consumer and investor preferences. For example, the adoption of renewable energy technologies, carbon pricing mechanisms, or stricter emission regulations can disrupt traditional business models and industries dependent on fossil fuels (Semieniuk et al., 2021).

Technological advancements and cost reductions in green technologies such as solar panels, electric vehicles, and carbon capture systems further influence transition dynamics. Moreover, growing public and investor pressure for environmentally responsible practices is leading companies to reconsider their long-term strategies (Campiglio et al., 2022). These factors can alter asset valuations, affect credit risks, and require firms to make significant adjustments in operations, which may result in financial volatility and market restructuring.

3.3.4. Climate change policies

Transition risks are deeply intertwined with the implementation of climate policies (Table 1). In pursuit of the Paris Agreement's goals, which aim to limit global warming to below 2.00°C above pre-industrial levels, governments need to introduce ambitious climate actions, including carbon pricing, regulatory mandates, subsidies for green energy, and emission reduction targets (Rogelj et al., 2016). However, the pace and scale of these policy implementations may also trigger unintended economic consequences. Central banks and financial regulators have expressed concern that sudden or uncoordinated policy shifts could lead to transition-induced economic disruptions, potentially contributing to recessionary pressures (Calice and Demekas, 2024). In response, financial authorities are exploring macroprudential policies that differentiate between green and brown assets, incentivizing environmentally sustainable investments while discouraging harmful investments (Grunewald, 2023). Establishing such frameworks in advance is crucial to mitigating systemic risk, enhancing the resilience of financial institutions, and preventing the rollback of climate commitments in the face of economic stress. For these efforts to be successful, climate policies and macroprudential regulations need to be viewed as complementary tools and integrated to facilitate a stable and sustainable economic transition (Carattini et al., 2023).

India is taking steps to reduce its carbon footprint. In 2008, the country introduced the National Action Plan on Climate Change (NAPCC), which outlines strategies for dealing with climate change and improving environmental sustainability. As part of the NAPCC, eight national missions (Fig. 4) were launched to address various aspects of development, adaptation, mitigation, and sustainability in India (Chateau et al., 2023).

Table 1

Comparative overview of key climate policies adopted across different regions around the world.

Climate policy	Year	Objective	Reference
Kyoto Protocol	1997	Reducing greenhouse gas (GHG) emissions	Occhipinti and Verona (2020)
Clean Development Mechanism	2001	Encouraging emission reduction projects in developing countries	Stahlke (2020)
European Union Emissions Trading System	2005	Creating a market for trading GHG emission allowances	Flachsland et al. (2020)
Copenhagen Accord	2009	Pledging to limit global temperature rise below 2.00°C	Maslin et al. (2023)
Cancun Agreements	2010	Addressing climate change through mitigation and adaptation actions	Di Pietro Paolo (2020)
Doha Amendment	2012	Strengthening Kyoto Protocol commitments	Rossati (2021)
Warsaw International Mechanism for Loss and Damage	2013	Addressing loss and damage associated with climate change impacts	Janzen et al. (2021)
Lima-Paris Action Agenda	2014	Mobilizing climate action at all levels	Jernnäs and Lövbrand (2022)
Paris Agreement	2015	Limiting global temperature rise below 2.00°C	Höhne et al. (2015)
Marrakech Partnership for Global Climate Action	2016	Enhancing climate action from various stakeholders	Bäckstrand and Kuyper (2024)
Talanoa Dialogue	2017	Facilitating open dialogue on climate action	Kirsch (2021)
Katowice Climate Package	2018	Finalizing guidelines for the Paris Agreement implementation	Havukainen and Kahiluoto (2023)
Madrid Climate Change Conference	2019	Addressing outstanding issues under the Paris Agreement	Streck (2020)
European Green Deal	2019	Achieving climate neutrality by 2050	Wolf et al. (2021)
United Kingdom's Net Zero Target by 2050	2019	Achieving net-zero GHG emissions by 2050	Dixon et al. (2022)
United States' Rejoining the Paris Agreement	2021	Reinstating commitment to the Paris Agreement	Kitetu and Ko (2022)
Conference of the Parties (COP) 26 Glasgow Climate Pact	2021	Accelerating global climate action and adaptation	Mountford et al. (2021)
COP 28 Dubai	2024	Transition away from fossil fuels to renewables such as wind and solar power	Icaza et al. (2024)



Fig. 4. National missions under the National Action Plan on Climate Change (NAPCC) in India.

3.4. Climate risk assessment and management

Climate change is causing noticeable effects on both the environment and human activities. This has prompted global researchers to concentrate on finding solutions for adapting to climate change, reducing the risk of disasters, managing natural resources, promoting sustainable farming practices, and embracing the circular economy. These efforts are particularly directed toward benefiting society (Schmidt-Thome, 2017; Day et al., 2019; Lara et al., 2022). The effect of climate change can be tackled by acting in disaster risk reduction, adopting sustainable farming methods, and embracing the circular economy (Iancu et al., 2022). Various policy tools can be utilized to encourage efforts in reducing the impact of climate change, whether at the national level or at a global scale. Each approach has its own advantages and disadvantages (Kirk and Cradock-Henry, 2022; Ndue and Pál, 2022). Regulation and standardization can bring clarity about emission levels, but they might also stifle innovation (Kaján et al., 2013). Figure 5 shows drivers, effects, mitigation strategies, and adaptation strategies of climate change.

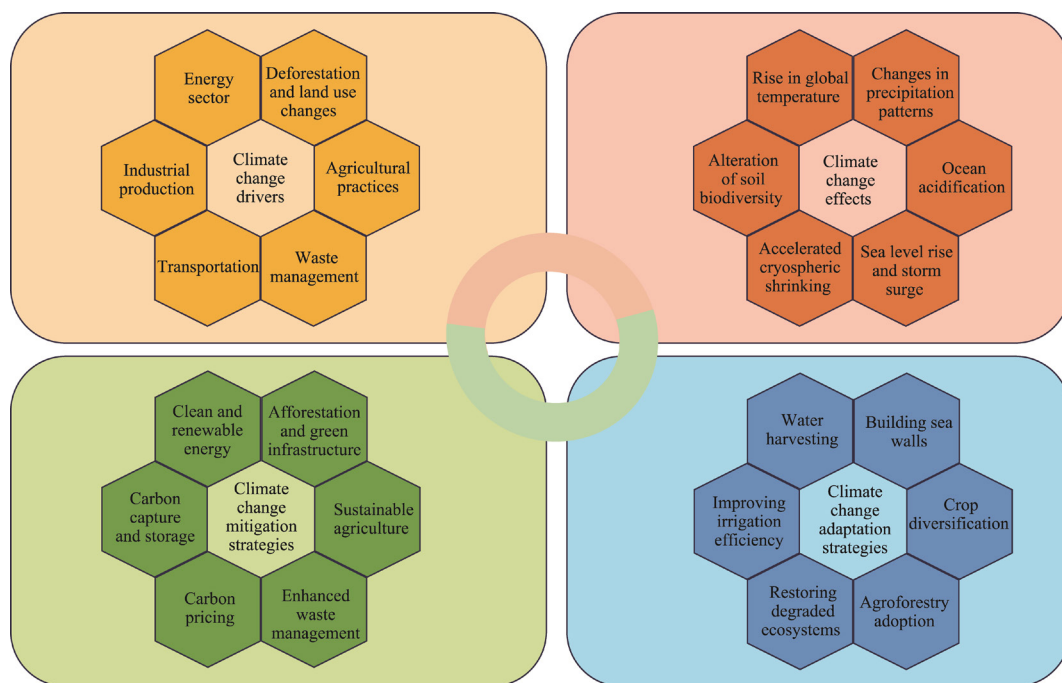


Fig. 5. Drivers, effects, mitigation strategies, and adaptation strategies of climate change.

Implementing carbon taxes can serve as a valuable incentive for mitigating climate change, but it doesn't guarantee a specific reduction. To address this challenge, subsidies and credits emerge as potential solutions, supporting the advancement and widespread adoption of crucial and expensive new technologies in society (Kaján et al., 2013; Schmidt-Thome, 2017; Day et al., 2019; Lara et al., 2022). Local or regional policies can pose challenges for companies and may impact people's livelihoods, but they are essential for adapting to climate change. Adequate funding for research plays a crucial role, as evident in numerous collaborations at the European level, driven by policies implemented by the European Commission (Asseng et al., 2019; Iancu et al., 2022; Rahman et al., 2022). Figure 6 depicts major mitigation-oriented policies across five most emission-intensive sectors in India.

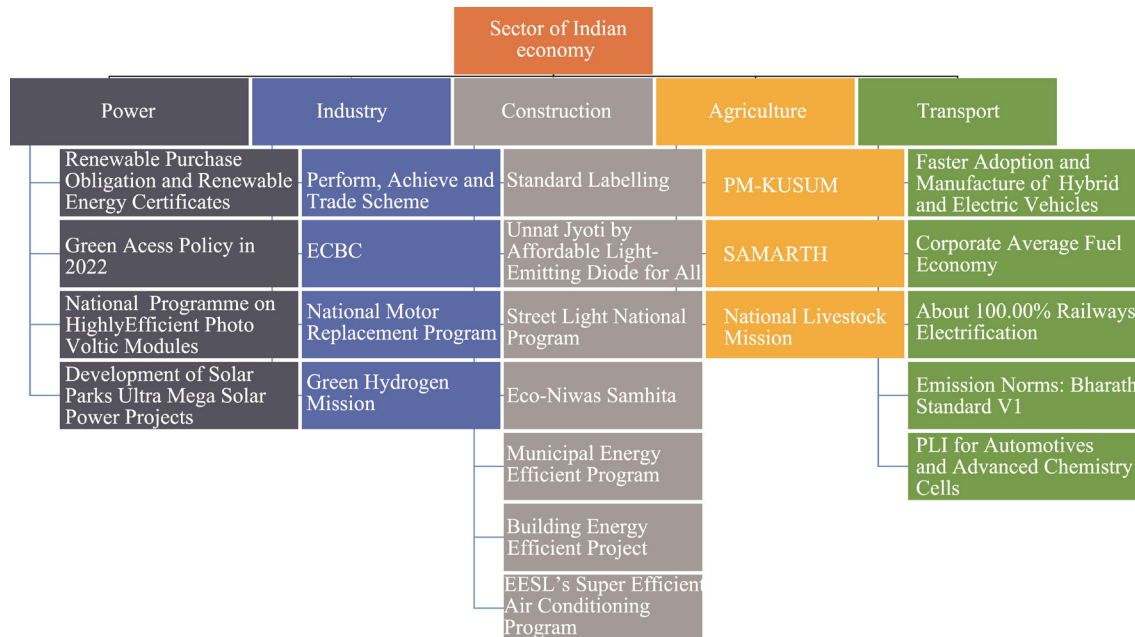


Fig. 6. Major mitigation-oriented policies across five most emission-intensive sectors in India. ECBC, Energy Conservation Building Code; PM-KUSUM, Pradhan Mantri Kisan Urja Suraksha evam Utthan Mahabhiyan; SAMARTH, Sustainable Agrarian Mission on Use of Agri-Residue in Thermal Power Plants; EESL, Energy Efficiency Services Limited; PLI, Production-Linked Incentive.

3.5. Climate Action Tracker

The Climate Action Tracker serves as a vital tool for monitoring and evaluating global efforts to combat climate change. It tracks the implementation of policies and agreements at multiple levels ranging from individual actions to international commitments and provides critical insights into the drivers influencing the success or failure of these climate initiatives (Tosun, 2022).

One of the major challenges identified in global climate governance is the lack of transparency, comparability, and comprehensiveness in climate finance tracking. Effective climate action hinges on the availability of accurate data regarding financial flows; however, current systems fall short in offering a unified framework (Buchner et al., 2011). To overcome this, scholars and policy-makers have called for a standardized, transparent framework for monitoring climate finance. This would enhance the credibility of reporting and foster greater accountability.

In response to these concerns, the Climate Action Tracker has been introduced. This automated analytical tool supports the systematic evaluation of public climate and energy policies across countries, offering a data-driven approach to monitor progress on Nationally Determined Contributions and climate action plans (CAT, 2019). Together, the Climate Action Tracker and related tools are critical in holding governments accountable and identifying gaps in climate mitigation efforts.

3.6. Assessment models of climate change

There is broad scientific consensus that climate change is real and presents serious risks on ecosystems, economies, and societies (IPCC, 2007). Scientists have developed General Circulation Models that offer increasingly sophisticated insights into the physical processes driving climate change. These models can be validated through back casting, that is, by comparing model projections with historical data to test accuracy (Ackerman et al., 2009). In addition to physical models, IAMs are widely used by economists to evaluate climate

policy options. IAMs integrate insights from climate science and economics, enabling researchers to explore the costs and benefits of different mitigation and adaptation strategies over time.

Until now, most policies and research mainly used the following five IAMs: (1) Dynamic Integrated Climate-Economy Model; (2) Climate Framework for Uncertainty, Negotiation, and Distribution; (3) Policy Analysis of the Greenhouse Effect; (4) Environmental Impact and Sustainability Applied General Equilibrium Model; and (5) Energy-Environment-Economy Macro Econometric Model. Dynamic Integrated Climate-Economy Model focuses on intertemporal optimization, selecting policy pathways that maximize social welfare over long-time frames (typically over a century) using nonlinear utility functions (Diaz and Moore, 2017; Grubb et al., 2021). Climate Framework for Uncertainty, Negotiation, and Distribution is extensively used in both academic and policy-making contexts. It simulates climate damage and adaptation across regions, often employing Monte Carlo simulations to account for uncertainty (Interagency Working Group on Social Cost of Carbon, 2010; Gillingham et al., 2015). Policy Analysis of the Greenhouse Effect assesses the cost-effectiveness of mitigation and adaptation strategies through probabilistic modeling and uncertainty analysis (Hope et al., 1993). Environmental Impact and Sustainability Applied General Equilibrium Model combines a computable general equilibrium economic framework with climate modules to assess the impacts of climate change on sectors and regions (Roson and van der Mensbrugghe, 2012). Energy-Environment-Economy Macro Econometric Model is a non-equilibrium simulation-driven model and incorporates feedback loops among technological diffusion, policy impacts, and macroeconomic dynamics, offering a more realistic evaluation of low-carbon transitions (Mercure et al., 2018).

IAMs have played a critical role in shaping the United States government's estimates of the social cost of carbon. However, the major limitation is their static treatment of economic growth. Many traditional IAMs do not adequately capture how climate change may directly influence GDP growth trajectories, particularly in developing economies. Study by Moore and Diaz (2015) has suggested that climate change could significantly depress growth rates in poorer countries, leading to more pressing needs for near-term mitigation. Emerging IAMs now seek to improve resolution and realism by integrating behavioral economics, technological spillovers, and international policy scenarios such as carbon pricing, emissions trading schemes, and cooperative agreements (Ciscar et al., 2012; Nordhaus, 2018).

4. Limitations and future research

Despite a surge in scholarly attention toward the economic impact of climate change, critical gaps remain in both the depth and breadth of current research. Most studies focus on aggregated national or global indicators, such as GDP and FDI, but fail to adequately capture sector-specific, subnational, and localized dynamics, particularly in developing countries, where vulnerability to climate-related risks is more acute. While IAMs have become foundational tools in forecasting long-term climate and economic outcomes, they often overlook key feedback loops between economic growth and climate shocks. Additionally, many IAMs underrepresent institutional heterogeneity and regional disparities, leading to generalized projections that may not effectively translate into localized policy contexts.

To address these challenges, future research should prioritize the development of high-resolution, regionally calibrated climate-economic models. These models need to account for local adaptation capacities, social vulnerability, and sectoral interdependency. Interdisciplinary integration linking behavioral economics, climate science, policy analysis, and econometrics is essential to construct more realistic and actionable frameworks. Furthermore, there is an urgent need to enhance the modeling of financial and systemic climate risks, integrate real-time climate data into economic forecasting, and expand the empirical use of the Climate Action Tracker.

5. Conclusions

Addressing climate change requires not only scientific innovation but also coherent policy frameworks, interdisciplinary expertise, and international cooperation. Establishing standardized climate risk assessment procedures and enhancing collaboration among experts allow for robust evaluation at various scales across countries, sectors, and institutional portfolios.

There is a need to consider short-term climate variability when communicating long-term risks to communities. Financial instruments play a vital role in softening the economic shocks and long-term costs associated with climate change. The potential for significant declines in real GDP per capita further highlights the urgency of immediate action. The impact of climate risks on investment and demand stresses their destabilizing potential for both financial

markets and supply chains. To counterbalance this, alternative economic models such as degrowth should be explored, especially for developing countries that are at risk of being marginalized during growth-oriented green transformation process.

In conclusion, climate change presents a complex matrix of risks and opportunities. As global temperatures rise and extreme weather events intensify, the importance of mitigation, adaptation, and resilience-building becomes more evident. The Paris Agreement offers a global framework, but the economic transition can be carefully managed to avoid unintended consequences, such as recession or increasing inequality. Robust evaluation mechanisms, including IAMs and tools like the Climate Action Tracker, are essential for measuring progress and guiding policy. Instruments such as carbon pricing and green subsidies can facilitate emission reductions, but only if they are supported by transparent monitoring and inclusive governance.

Looking ahead, the path to a sustainable and prosperous future lies in enacting reliable and flexible climate legislation, innovating economic frameworks that balance growth with sustainability, and deepening international collaboration. Future research can deepen the understanding of long-term macroeconomic effects, especially in economically uncertain regions, and support the creation of ethical, inclusive, and efficient legal frameworks that champion both economic resilience and environmental stewardship.

Authorship contribution statement

Mahalingam Santhosh KUMAR: conceptualization, data curation, formal analysis, methodology, visualization, writing - original draft, and writing - review & editing; Narasimabhrathi Venkatesa PALANICHAMY: supervision; K. M. SHIVAKUMAR: investigation, methodology, and validation; Mani CHANDRAKUMAR: methodology, supervision, and validation; Muthuswamy KALPANA: investigation, software, and validation; and Dhandapani MURUGANANTHI: conceptualization, formal analysis, and supervision. All authors approved the manuscript.

Declaration of conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this study.

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